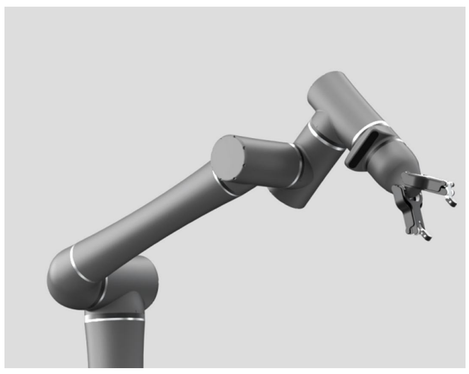
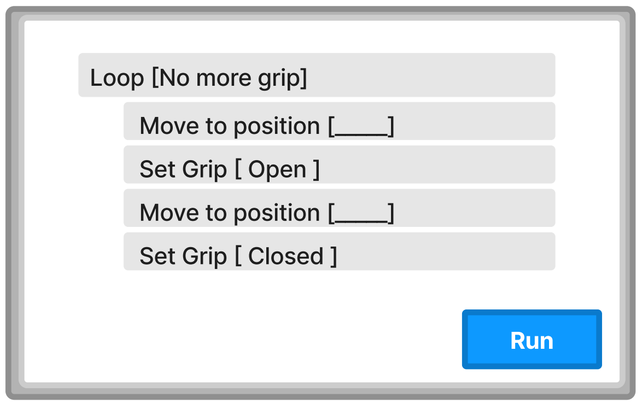


Task

- Architect system which allows users to run a routine
- UX already exists, you design “what happens when the user clicks Run”

UX



R01 Robot

HW

- Full Linux machine onboard: ~equivalent specs to mac mini m1
- RGB Camera
- Power supply + WiFi/Ethernet
- iPad - not physically connected to arm

Sensors

- 6 joints - each emit rotational position @ 1000HZ
- Camera sensor - 30FPS
- 1 gripper - emits grip size @ 100HZ

Actuators

- 6 joints - can send velocity to joints - max 1000HZ
- Gripper - can send grip size commands - max 100HZ

IPad / UX / Front End app (web front end)

Server program (running on robot) - exposes HTTP / Web sockets , commands

Functional requirements:

Evaluate conditions that require robot state (ex: is gripper open)

Commands to move the robot, gripper

Safety / obstacle detection and avoidance

Using sensor to reason about objects to grasp / finding places to move

Command:

- name (method)
- arguments and data types
- return type

Command tree/graph:

- A graph of command objects with their dependencies, parent-child relation ships

Version management:

High level system components:
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1. Control loop running on the robot (omnipresent thread) runs at a control frequency
2. HTTP server component which exposes commands, takes requests to execute commands, read requests like state and other interesting things
3. ~~UX app calling the HTTP end point with the requests~~
4. Perception component for safety / obstacle avoidance and object and grasp pose detection
5. Sending the whole program as a command graph to the server and executing it

Perception:

-

Planning:

- outputs a trajectory (positions or velocities)

Move to position:

- IK to find target joint angles
- Send angles to the control (position control)

Gripper:

Control loops: executes trajectory using appropriate control method

Camera:

- emits images messages at 30 GPS

Perception system:

- subscribes to camera messages
- Runs perception models outputs object poses, grasp poses and emits them as messages

Planning:

- subscribe to perception messages
- keeps the state of running user program

Control loop:

- Takes commands from planning and executes the control method

Questions:

- Node structure / break up
- Message passing and receiving at different rates
- Message failure, dedup, buffer sizes